Pilatus PC-12/45 (PC12/45)

Pratt & Whitney PT6A-67B, 1,200 shp (flat-rated) · 1–2 crew + 6 to 9 seats · Typical legacy PC-12/45 (1996–2005, PT6A-67B, 9,920 lb, 5.75 psi cabin, Bendix/King analogue panel with EFIS-40 and KFC 325, hydraulic gear) AFM procedures; no onboard electronic checklist. Battery 2 on later serials. Torque in psi as displayed; limits marked 'verify' should be checked against the aircraft's AFM.

V-speeds & limits

Vr	**80–85 KIAS (flaps 15, by weight)**
Vx	**110 KIAS**
Vy	**120 KIAS (flaps UP)**
Best glide	**approx 110–115 KIAS, prop feathered (approx 2.5 nm per 1,000 ft)**
Vfe	**165 / 130 / 120 KIAS (15° / 30° / 40°)**
Vlo	**180 KIAS extend / 180 KIAS retract**
Vle	**180 KIAS**
Va	**165 KIAS (9,920 lb)**
Vmo	**236 KIAS**
Vs / Vso	**approx 85–90 / 64–68 KIAS (9,920 lb, flaps 0 / 40 – verify)**
Vref	**80–90 KIAS (flaps 40, by weight; approx +10 KIAS flaps 15)**
Min icing speed	**approx 130 KIAS flaps UP (verify AFM placard)**

Normal procedures

Before Start

- Preflight inspection — **COMPLETE (covers, engine inlet plugs, chocks removed; inertial separator / gear doors / boots inspected)**
- Weight & balance / takeoff data — **COMPUTED (flap 15, Vr 80–85, Vx 110, Vy 120, torque 44.3 psi, field length)**
- Passenger briefing — **COMPLETE (belts, airstair / cargo door, overwing exit, oxygen drop-out masks, no smoking)**
- Seats, belts, shoulder harnesses — **ADJUST & LOCK**
- Control lock — **REMOVED & STOWED**
- Cabin door / cargo door — **CLOSED & LOCKED (handles flush, CAWS door lights out)**
- Parking brake — **SET**
- Landing gear selector — **DOWN**
- Power control lever (PCL) — **IDLE (out of beta / reverse)**
- Manual override (MOR) lever — **FULL AFT (OFF / stowed)**
- Condition lever — **CUTOFF**
- Fuel emergency shutoff (firewall valve) — **OPEN (guard closed)**
- Cabin pressurisation — **NORMAL (not DUMP); ECS / bleed air OFF for start**
- Avionics 1 / 2, inverters — **OFF**
- Generator 1 / Generator 2 switches — **ON**
- Bus tie — **NORMAL / closed**
- Battery 1 (and Battery 2 if fitted) — **ON – voltage CHECK (approx 24–25 V; 24 V minimum for a battery start – verify)**
- CAWS — **TEST (all annunciators, master warning / caution, aural tones) – then LAMP TEST**
- Annunciators — **CHECK – only expected lights (GEN 1 / GEN 2 OFF, INVERTER, PROP LOW PITCH etc.)**
- Fuel quantity — **CHECK L / R (approx 201 US gal usable per wing), balanced within 26 US gal**
- Fuel pumps L / R — **AUTO**
- Oxygen — **CHECK (bottle pressure, crew masks, passenger drop-out AUTO)**
- Trims (pitch / rudder / aileron) — **CHECK indicators – SET for takeoff**
- Beacon — **ON**

Engine Start

▶ Power source for start?

Battery:

Battery start: 24 V minimum. A weak battery means a hot, slow start – abort if Ng hangs below 13% or ITT rises rapidly.

External power:

External power ..... **CONNECTED – EXT PWR switch ON, EXT PWR annunciator, bus approx 28 V**

Propeller area ..... **CLEAR**

Ignition ..... **AUTO**

Fuel pumps L / R ..... **AUTO (L / R FUEL PUMP advisories ON for start)**

Starter ..... **ON – STARTER annunciator ON; limits 30 sec ON / 1 min OFF, 30 min cool after 3 cycles**

Oil pressure ..... **RISING**

Ng ..... **13% MINIMUM, stable**

Condition lever ..... **GROUND IDLE**

ITT / Ng ..... **LIGHT-OFF within 10 sec, Ng accelerating; MONITOR ITT (1,000°C max for 5 sec)**

Abort start if no light-off within 10 sec, ITT approaches 1,000°C, or Ng hangs: condition lever CUTOFF, keep motoring until ITT falls, then starter OFF.

Starter ..... **AUTOMATIC CUT-OUT at approx 50% Ng (STARTER light out) – if not, starter OFF**

Ng – ground idle ..... **STABILISED (approx 60–66% – verify)**

Oil pressure ..... **CHECK (green; 90–135 psi normal, 60 psi min at idle – verify)**

Ignition ..... **AUTO (IGNITION light out)**

Generator 1 ..... **ON LINE – GEN 1 OFF caution out, load / volts CHECK**

Generator 2 ..... **ON LINE – GEN 2 OFF caution out, bus tie closed, approx 28 V**

External power ..... **OFF & DISCONNECTED (if used; EXT PWR light out)**

Inverter 1 / 2 ..... **ON – INVERTER caution out; select each in turn, leave one ON**

Avionics 1 / 2 ..... **ON**

ECS / bleed air ..... **ON (as required for cabin heating / cooling)**

Fuel pumps L / R ..... **AUTO (FUEL PUMP advisories out, FUEL PRESS out)**

Engine instruments / CAWS ..... **CHECK (ITT, torque, Ng, Np approx 1,000 RPM, oil pressure / temperature, fuel flow; no red lights)**

Before Taxi

EFIS-40 / flight instruments / avionics ..... **CHECK & SET (ADI / EHSI no flags, standby instruments, altimeters SET, nav / com / GPS loaded)**

Stall warning / stick pusher ..... **TEST – hold test switch: shaker, then pusher pulse (column moves forward); PUSHER light sequence correct**

Pusher interrupt (control wheel) ..... **PRESS & HOLD during test – pusher interrupted**

Electric trims ..... **TEST (pitch both halves of split switch, rudder, aileron) – trim interrupt / A/P disconnect stops motion; SET for takeoff**

Autopilot (KFC 325) ..... **PREFLIGHT TEST complete (PFT) – ENGAGE, check disconnect (wheel button, trim); OFF**

Yaw damper ..... **OFF**

Pitot / static heat 1 & 2, stall warning / probe heat ..... **ON momentarily – cautions OUT, ammeter rise; then AS REQUIRED**

Propeller deice ..... **ON – PROP DE ICE light / ammeter CHECK; then AS REQUIRED**

Deice boots ..... **1 CYCLE – boots inflate / deflate, BOOTS lights; then OFF**

Windshield heat L / R ..... **ON – CHECK; then AS REQUIRED**

Inertial separator ..... **OPEN – INERT SEP OPEN advisory (approx 15 sec); then AS REQUIRED (OPEN for icing, contaminated or gravel surfaces)**

Ice light ..... **CHECK (night / icing)**

Flaps ..... **CYCLE 0 – 15 – 30 – 40, indicator CHECK; SET 15 for takeoff**

Cabin pressurisation controller ..... **SET (cruise altitude, rate as required); cabin NORMAL**

Brakes ..... **CHECK on initial roll (PCL into beta for speed control)**

Flight instruments ..... **CHECK in turns (EHSI / compass, turn coordinator)**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Propeller overspeed governor	**TEST – hold PROP OVSP TEST, PCL advance: Np stabilises below 1,700 RPM at the test value (approx 1,500–1,600 RPM – verify); release, PCL IDLE**
Manual override (MOR)	**CHECK – advance slowly: Ng / ITT increase, then FULL AFT; PCL IDLE**
Engine instruments	**CHECK (oil temp in green, oil pressure 90–135 psi, ITT, Ng, torque, fuel flow)**
Fuel	**CHECK quantity / balance, pumps AUTO, FUEL PRESS out, emergency shutoff OPEN**
Electrical	**CHECK (GEN 1 / GEN 2 on line, loads within limits, bus tie closed, approx 28 V, inverter ON)**
Pressurisation / ECS	**Bleed air ON, ECS AUTO, cabin NORMAL, controller SET**
Condition lever	**FLIGHT IDLE**
Flaps	**15 – indicating**
Trims	**SET for takeoff (pitch per loading; rudder / aileron neutral)**
Ice protection	**AS REQUIRED (inertial separator OPEN below 5°C in visible moisture / contaminated runway – reduced torque available; pitot / static / probe heat ON)**
Yaw damper / autopilot	**OFF**
Oxygen	**CREW MASKS available; passenger system AUTO**
Lights	**AS REQUIRED**
Transponder	**ALT**
CAWS / doors	**NO RED OR AMBER; CABIN DOOR / CARGO DOOR lights out**
Takeoff briefing	**COMPLETE (abort before Vr 80–85; engine failure after takeoff: condition lever CUTOFF – feather, flaps, land ahead; limits 44.3 psi / 800°C / 1,700 RPM)**

After Takeoff / Climb

Landing gear	**UP (positive rate, below 180 KIAS) – 3 green out, gear lights out, HYDR pump stops**
Flaps	**UP (above 110 KIAS, clear of obstacles)**
Climb power	**SET (torque 36.9 psi max climb, ITT 760°C max, Np 1,700 RPM – verify)**
Airspeed	**120 KIAS (Vy) / 140–160 KIAS cruise climb**
Yaw damper	**ON**
Autopilot	**AS REQUIRED**
Pressurisation	**CHECK (cabin climbing per rate, differential increasing, 5.75 psi max)**
Ice protection / inertial separator	**AS REQUIRED (CLOSED when clear of icing / contaminated surfaces)**
Fuel pumps	**AUTO – FUEL PUMP advisories out**
Engine instruments / lights	**CHECK; landing / taxi lights OFF**

Cruise

Cruise power	**SET (torque per chart, 36.9 psi max, ITT 760°C max, 1,700 RPM)**
Fuel	**CHECK quantity L / R; automatic balancing (heavy-side pump runs); imbalance within 26 US gal – balance manually with heavy-side pump ON if required**
Pressurisation	**CHECK (cabin altitude, rate, differential 5.75 psi max; CAB PRESS warning at 10,000 ft cabin)**
Oxygen	**CHECK (pressure, masks available)**
Engine instruments	**CHECK (oil 90–135 psi / temperature in green, ITT, Ng 104% max)**
Electrical	**CHECK (both generators on line, loads balanced, 28 V)**
Ice protection	**AS REQUIRED (inertial separator OPEN before entering icing; approx 130 KIAS minimum in icing; pusher ICE MODE with boots selected)**
Altimeters / OAT	**CHECK; cruise altitude vs cabin**

Descent

Altimeters	**SET**
Cabin pressurisation controller	**SET landing field elevation (rate approx 500 fpm or as required)**
ECS / bleed air	**ON – cabin temperature as required**
Ice protection / inertial separator	**AS REQUIRED (inertial separator OPEN for icing or contaminated runway)**
Windshield heat	**AS REQUIRED**
Fuel	**CHECK quantity / balance, pumps AUTO**
Approach briefing	**COMPLETE (Vref 80–90 flaps 40; flaps 15 / 30 / 40 below 165 / 130 / 120; gear below 180; missed approach)**
Seat belts / passenger briefing	**SECURE**

Before Landing

Pressurisation	**CHECK (cabin at or above field elevation, differential approaching zero)**
Flaps	**15 (below 165 KIAS)**
Landing gear	**DOWN (below 180 KIAS) – 3 GREEN, gear unsafe light out, HYDR pump stops**
Yaw damper / autopilot	**OFF**
Condition lever	**FLIGHT IDLE**
Fuel pumps	**AUTO**
Ice protection	**AS REQUIRED (inertial separator OPEN for contaminated runway)**
Flaps	**30 (below 130 KIAS), then 40 (below 120 KIAS) when landing assured**
Airspeed	**Vref 80–90 KIAS flaps 40 (by weight); add approx 10 KIAS flaps 15**
Landing lights	**ON**
Landing gear	**3 GREEN – CONFIRMED on final**

After Landing

PCL	**BETA / REVERSE as required (lift gate) – out of reverse by approx 40 KIAS**
Flaps	**UP**
Condition lever	**GROUND IDLE**
Trims	**RESET for takeoff**
Ice protection	**OFF (pitot / static / probe heat, prop deice, boots, windshield heat)**
Inertial separator	**AS REQUIRED (OPEN for taxi on contaminated / gravel surfaces)**
Transponder / lights	**STBY / AS REQUIRED**

Shutdown

Parking brake	**SET**
Avionics 1 / 2	**OFF**
ECS / bleed air	**OFF**
Ice protection / electrical load	**OFF**
Inverters	**OFF**
ITT	**STABILISED – 1 minute at GROUND IDLE**
Condition lever	**CUTOFF (propeller feathers during run-down)**
ITT / Ng	**MONITOR during run-down (no ITT rise)**
Fuel pumps L / R	**OFF**
Oxygen	**OFF**
Exterior lights / beacon	**OFF**
Battery 1 / 2	**OFF (after propeller stopped)**
Control lock	**INSTALL**
Inlet plugs / covers / chocks	**AS REQUIRED**

Abnormal procedures

Generator Failure (GEN 1 / GEN 2 OFF caution)

GEN 1 or GEN 2 caution, generator load zero

Affected generator switch	**OFF, then RESET, then ON**

▶ Generator restored?

Yes:

Loads / voltage	**MONITOR both generators**

No – one generator failed:

Affected generator switch	**OFF**
Electrical load	**REDUCE within remaining generator limit (Gen 2 is the smaller unit – shed heavy loads: windshield heat, prop deice, ECS as required)**
Bus tie	**CHECK closed – all buses powered**
Deferred: One generator inoperative – electrical load limited, land as soon as practical.	**NOTED**

No – both generators failed:

Both generator switches	**OFF**
Non-essential bus / Avionics 2	**OFF (battery only – approx 30 min endurance, less with Battery 1 only)**
Essential bus	**CHECK powered (ESNTL BUS warning out) – essential avionics only**
ECS / bleed air	**Cabin heating / pressurisation reduced; descend as required**
Landing gear	**PLAN early extension while battery power remains; emergency hand pump available**
Flaps	**SELECT early – electric; may be slow or unavailable**
Deferred: Both generators failed – battery only; extend gear and flaps early, land as soon as possible.	**NOTED**

Inverter Failure (INVERTER caution)

Inverter switch ..... **SELECT other inverter (1 / 2)**

▶ INVERTER caution out?

Yes:

- AC instruments ..... **CHECK (AC-powered flight instruments / EFIS-40 symbol generator)**
- Deferred: One inverter inoperative – single AC source, maintenance required. ..... **NOTED**

No – both failed:

- Inverters ..... **OFF**
- Flight instruments ..... **USE standby attitude / altimeter / airspeed and EFIS reversionary as available**
- AC-powered avionics ..... **NOTE inoperative; autopilot may be unavailable**
- Deferred: Both inverters failed – AC-powered instruments / autopilot inoperative, standby instruments only. ..... **NOTED**

Low Oil Pressure / Chip Detector

OIL PRESS warning / oil pressure below 90 psi above idle, OIL TEMP, or CHIP caution

▶ Indication?

Oil pressure low:

- Oil pressure / temperature ..... **CROSS-CHECK (60–90 psi at idle only; below 60 psi red – verify)**

▶ Pressure below 60 psi or temperature rising?

Yes:

- Power ..... **REDUCE to minimum required for flight**
- Landing ..... **NEAREST SUITABLE AIRPORT – expect engine seizure / failure; brief Engine Failure in Flight**
- Deferred: Low oil pressure – land as soon as possible, forced landing possible. ..... **NOTED**

No – 60–90 psi:

- Power ..... **REDUCE; monitor pressure and temperature**
- Deferred: Oil pressure in caution range – land as soon as practical, maintenance check. ..... **NOTED**

CHIP caution:

- Engine instruments ..... **MONITOR oil pressure / temperature, vibration, torque**
- Power ..... **REDUCE to minimum practical**
- Landing ..... **NEAREST SUITABLE AIRPORT – shut down and feather only if other indications confirm failure**
- Deferred: Chip detector – land as soon as possible, engine inspection required. ..... **NOTED**

Fuel Imbalance / Fuel Pump Failure / FUEL PRESS Low

FUEL PRESS caution, L / R FUEL PUMP fail, imbalance above 26 US gal, or L / R FUEL LOW

▶ Indication?

Fuel imbalance:

- Fuel pump (heavy side) ..... **ON (manual balancing – engine draws from the pumped side)**
- Fuel pump (light side) ..... **AUTO / OFF**
- Fuel quantity ..... **MONITOR – pumps AUTO when balanced; max imbalance 26 US gal (verify)**

▶ Imbalance correctable?

Yes:

- Deferred: Automatic fuel balancing inoperative – manual balancing with fuel pumps, maintenance check. ..... **NOTED**

No:

- Landing ..... **PLAN with imbalance – aileron trim, into-wind wing heavy side if possible, longer runway**
- Deferred: Fuel imbalance above limit – lateral trim, gentle manoeuvring, land as soon as practical. ..... **NOTED**

FUEL PRESS caution:

- Fuel pumps L / R ..... **ON (both)**
- Fuel quantity ..... **CHECK both sides; emergency shutoff OPEN**

▶ FUEL PRESS caution out?

Yes:

- Fuel pumps ..... **LEAVE ON**
- Deferred: Engine-driven fuel pump suspect – booster pumps on, land as soon as practical. ..... **NOTED**

No:

- Power ..... **REDUCE; expect fuel starvation – brief Engine Failure in Flight**
- Deferred: Fuel pressure low with pumps on – land as soon as possible. ..... **NOTED**

L / R FUEL PUMP fail or FUEL LOW:

- Other side fuel pump ..... **ON to feed from the side with a serviceable pump**
- Fuel quantity ..... **MONITOR (FUEL LOW approx 20 US gal per side – verify); avoid uncoordinated flight**
- Deferred: Fuel pump / low fuel – land as soon as practical, fuel from one side only. ..... **NOTED**

Pressurisation Fault / Cabin Not Holding

Cabin altitude or rate not following the controller, differential abnormal, CAB PRESS warning

Cabin pressurisation switch ..... **NORMAL (not DUMP)**

Bleed air / ECS ..... **ON / AUTO**

Controller ..... **RESELECT cabin altitude / rate**

Ng ..... **INCREASE if low in descent (more bleed air)**

Cabin differential ..... **CHECK – 5.75 psi max (safety valve relieves above)**

▶ Cabin response?

Cabin following again:

Pressurisation ..... **MONITOR for remainder of flight**

Deferred: Pressurisation controller intermittent – maintenance check. ..... **NOTED**

Cabin climbing:

Oxygen ..... **CREW MASKS ON if cabin above 10,000 ft; passenger oxygen as required**

Altitude ..... **DESCEND to keep cabin below 10,000 ft – Rapid Decompression procedure if CAB PRESS warning**

Deferred: Pressurisation failed – remain at or below 10,000 ft, oxygen as required. ..... **NOTED**

Differential too high:

Altitude ..... **DESCEND – reduce differential**

Bleed air ..... **OFF briefly to reduce inflow**

Before landing ..... **DUMP below 10,000 ft cabin so the cabin is depressurised at touchdown**

Deferred: Outflow valve fault – depressurise with DUMP before landing; do not open doors until differential is zero. ..... **NOTED**

Bleed Air Leak / Overtemperature (BLEED AIR / ECS caution)

BLEED AIR caution, duct overheat, or hot air / burning smell from vents

Bleed air ..... **OFF**

ECS ..... **OFF**

Cabin pressurisation ..... **MONITOR – cabin will climb; descend to keep below 10,000 ft**

Oxygen ..... **AS REQUIRED**

▶ Caution out and no fumes?

Yes:

Bleed air ..... **May be re-selected ON after cooling if cabin pressure needed; monitor**

Deferred: Bleed air fault – limited pressurisation, lower cruise altitude, maintenance required. ..... **NOTED**

No:

Bleed air / ECS ..... **LEAVE OFF for the remainder of the flight**

Smoke / fumes ..... **Use Smoke / Fumes procedure if required**

Deferred: Bleed air off – unpressurised, remain at or below 10,000 ft, no cabin heat. ..... **NOTED**

Ice Protection Failure (Inertial Separator / Boots / Prop Deice)

INERT SEP fail, BOOTS caution, PROP DE ICE caution, or ice accumulating with system selected

Icing conditions ..... **EXIT – climb / descend / turn**

Airspeed ..... **approx 130 KIAS minimum flaps UP while ice remains**

▶ Failed system?

Inertial separator:

Inertial separator switch ..... **RECYCLE once (approx 15 sec travel)**

Icing conditions ..... **AVOID – engine inlet unprotected; expect ITT rise / torque loss with ice ingestion**

Ignition ..... **ON in precipitation / icing (flame-out protection)**

Deferred: Inertial separator failed – avoid icing / contaminated runways, ignition ON in precipitation. ..... **NOTED**

Deice boots:

Boots switch ..... **OFF, then 1 CYCLE; check BOOTS lights**

Icing conditions ..... **EXIT – airframe ice will accumulate; pusher ICE MODE may not be active**

Landing ..... **FLAPS 15, Vref +10–15 KIAS, no abrupt pitch changes**

Deferred: Residual airframe ice – flaps 15 landing, Vref +10–15 KIAS, longer landing distance. ..... **NOTED**

Propeller deice:

Prop deice switch ..... **OFF then ON; circuit breaker CHECK**

Icing conditions ..... **EXIT – expect vibration from asymmetric ice; vary Np / power briefly to shed ice**

Deferred: Prop deice inoperative – avoid icing, expect vibration, maintenance required. ..... **NOTED**

Pitot / Static Heat Failure

PITOT 1 / 2 or STATIC HEAT caution with switch ON, or erratic airspeed / altitude in icing

Pitot / static heat switches ..... **CHECK ON**

Pitot / static heat circuit breakers ..... **CHECK / RESET once**

▶ Caution out?

Yes:

Flight ..... **CONTINUE, monitor**

No:

Icing conditions ..... **AVOID / EXIT**

Airspeed / altitude ..... **CROSS-CHECK pilot vs copilot vs GPS ground speed; fly by attitude and power if unreliable**

Alternate static source ..... **SELECT if static blockage suspected (altimeter / VSI / airspeed erratic)**

Deferred: Pitot / static heat failed – use opposite side indications, avoid icing, approach by attitude and power if unreliable. ..... **NOTED**

CAWS Caution Handling / CAWS Failure

Master caution with an amber light, or CAWS not responding to test

Master caution ..... **PRESS to reset / acknowledge**

Annunciator ..... **IDENTIFY – confirm with the relevant gauge / switch position before acting**

▶ Situation?

Single caution confirmed:

Procedure ..... **COMPLETE the matching abnormal checklist**

Caution with normal indications:

Lamp test ..... **PRESS – verify the annunciator and sensor; treat as spurious only with a confirmed normal gauge**

System ..... **MONITOR closely for the remainder of the flight**

Deferred: Spurious / unconfirmed caution – monitor system directly, maintenance check. ..... **NOTED**

CAWS inoperative (no test response):

Engine / system instruments ..... **MONITOR directly – no warning or caution protection (oil pressure, cabin altitude, fuel, gear)**

Altitude ..... **CONSIDER 10,000 ft or below (no cabin altitude warning)**

Deferred: CAWS inoperative – no annunciator protection, monitor gauges directly, maintenance required. ..... **NOTED**

Hydraulic Pressure Low / Gear Unsafe

HYDR caution (pump running continuously or low pressure), gear unsafe light, or fewer than 3 green

Airspeed ..... **BELOW 180 KIAS**

Landing gear selector ..... **CONFIRM position; indicator bulbs CHECK (lamp test)**

▶ Situation?

HYDR caution, gear up:

Landing gear ..... **PLAN emergency extension by hand pump (Landing Gear Emergency Extension) – do not rely on the electric pump**

Deferred: Hydraulic fault – emergency gear extension, no retraction after extension. ..... **NOTED**

HYDR caution, gear down:

Landing gear ..... **LEAVE DOWN – do not retract**

Airspeed ..... **180 KIAS maximum**

Deferred: Gear down for remainder of flight – 180 KIAS max, increased fuel burn. ..... **NOTED**

Gear unsafe after selection:

Landing gear selector ..... **RECYCLE once (UP then DOWN, below 180 KIAS)**

Gear warning ..... **CHECK – horn with PCL retarded and gear not down**

▶ 3 green?

Yes:

Deferred: Gear indication fault – do not retract after landing, maintenance inspection required. ..... **NOTED**

No:

Use Landing Gear Emergency Extension; if still unsafe, Gear-Up / Gear Unsafe Landing.

Flap Asymmetry / Flap Failure (FLAPS caution)

FLAPS caution (asymmetry protection has stopped the flaps), flaps not moving, or roll with flap selection

Flap selector ..... **RETURN to the current indicated position – do not cycle**

Roll ..... **COUNTER with aileron / trim; airspeed below Vfe for the extended flap (165 / 130 / 120 KIAS)**

Flap circuit breaker ..... **CHECK – reset once only if no asymmetry**

▶ Flaps?

Asymmetric – flaps stopped:

Flaps ..... **LEAVE as set for landing**

Approach speed ..... **Vref for the lesser flap +10 KIAS; longer runway**

Deferred: Flap asymmetry – flaps frozen, increased approach speed and landing distance, lateral trim required. ..... **NOTED**

Failed UP / 15 – symmetric:

Landing ..... **FLAPS UP: approx 100–105 KIAS approach; FLAPS 15: Vref +10 KIAS; longest runway**

Deferred: Flaps inoperative – flaps-up / 15 landing, increased landing distance. ..... **NOTED**

Operable after reset:

Flaps ..... **USE normally; monitor indicator for symmetry**

Deferred: Flap system reset in flight – maintenance check. ..... **NOTED**

Yaw Damper / Autopilot Failure (KFC 325)

A/P TRIM warning, autopilot disconnect, yaw damper off uncommanded, or abnormal autopilot behaviour

Autopilot / yaw damper ..... **DISCONNECT (wheel button) – fly manually**

Trim ..... **CHECK – retrim manually; electric trim test if A/P TRIM warning (Trim Runaway if moving uncommanded)**

▶ **Re-engagement after test?**

Normal – re-engaged:

Autopilot ..... **MONITOR closely; disconnect at the first anomaly**

Autopilot unavailable:

Flight ..... **HAND-FLY; reduce workload (radar vectors, simpler approach)**

Yaw damper ..... **OFF – expect yaw / Dutch roll tendency at altitude; use rudder trim**

Deferred: Autopilot / yaw damper inoperative – hand-flown approach, no coupled approach. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

On the runway, below Vr 80–85 KIAS

PCL (memory) ..... **IDLE, then BETA / REVERSE as available**

Brakes (memory) ..... **MAXIMUM – stop straight ahead**

Condition lever ..... **CUTOFF (if stopping / fire suspected)**

Fuel emergency shutoff ..... **CLOSED (if fire suspected)**

Battery 1 / 2 ..... **OFF after stop (if evacuating)**

Reverse is unavailable with the engine failed – plan on brakes only. Always stop below Vr.

Engine Failure After Takeoff

Airborne, insufficient runway remaining / altitude for a restart

LAND IMMEDIATELY

Condition lever (memory) ..... **CUTOFF (propeller FEATHERS)**

Airspeed (memory) ..... **110 KIAS flaps UP / 95–100 KIAS flaps 15 (minimum Vref +10)**

Flaps (memory) ..... **AS REQUIRED for the landing area (40 on short final if possible)**

Landing gear (memory) ..... **AS REQUIRED (DOWN for a prepared surface; UP for soft ground / water)**

Land (memory) ..... **STRAIGHT AHEAD – small heading changes only**

Fuel emergency shutoff ..... **CLOSED**

Fuel pumps ..... **OFF**

Cabin pressurisation ..... **DUMP**

Battery 1 / 2 / generators ..... **OFF before touchdown (when no longer required)**

Doors ..... **UNLATCHED before touchdown; brace passengers**

Do not attempt a turn back at low altitude. Feathered glide: approx 2.5 nm per 1,000 ft.

Engine Failure in Flight

Loss of power with altitude available (torque / Ng / ITT dropping, PROP LOW PITCH)

LAND AS SOON AS POSSIBLE

- Airspeed (memory) — **110–115 KIAS (best glide, approx 2.5 nm per 1,000 ft)**
- PCL (memory) — **IDLE**
- Condition lever (memory) — **CUTOFF (propeller FEATHERS)**
- Landing site (memory) — **SELECT – turn toward it; ATC advise**
- Cause — **CHECK (fuel quantity, FUEL PRESS, fuel pumps ON, emergency shutoff OPEN, ignition AUTO)**

▶ Restart practicable (no fire / mechanical damage, altitude and time available)?

Yes – starter-assisted air start:

- Airspeed / altitude — **WITHIN air start envelope (below approx 20,000 ft, 100–180 KIAS – verify)**
- Electrical load — **REDUCE (ECS / bleed OFF, non-essential OFF – battery only)**
- Condition lever — **CUTOFF – PCL IDLE**
- Fuel emergency shutoff — **OPEN; fuel pumps ON**
- Ignition — **ON**
- Starter — **ON (observe limits: 30 sec ON / 1 min OFF)**
- Ng — **13% MINIMUM**
- Condition lever — **GROUND IDLE**
- ITT — **LIGHT-OFF within 10 sec; MONITOR (1,000°C max 5 sec) – abort to CUTOFF if exceeded**
- Starter — **CUT-OUT at approx 50% Ng (STARTER light out)**

▶ Engine restarted?

Yes:

- Condition lever — **FLIGHT IDLE after Ng stable; ignition AUTO**
- Oil pressure / ITT / torque — **CHECK – power AS REQUIRED (gently)**
- Generators / bleed air / fuel pumps — **CHECK on line / AUTO**
- Deferred: Engine restarted after in-flight failure – land as soon as possible, maintenance investigation required. — **NOTED**

No:

- Condition lever — **CUTOFF; starter OFF; ignition OFF**

Do not attempt further starts. Continue with the forced landing.

No – forced landing:

Forced landing without power (propeller feathered):

- Passengers — **BRIEF – belts tight, brace; ATC / 7700**
- Cabin pressurisation — **DUMP (equalise before touchdown)**
- Fuel emergency shutoff — **CLOSED; fuel pumps OFF**

▶ Landing surface?

Prepared / firm – gear DOWN:

- Landing gear — **DOWN (below 180 KIAS; emergency hand pump if hydraulic pump unpowered – approx 25–40 strokes)**
- Flaps — **15 downwind, 40 on final when landing assured**
- Airspeed — **approx 100 KIAS on base, Vref 85–90 KIAS on final**

Soft / water / unknown – gear UP:

- Landing gear — **UP**
- Flaps — **40 on final**
- Airspeed — **Vref 85–90 KIAS; minimum sink at touchdown**
- Battery 1 / 2 / generators — **OFF before touchdown (when no longer required)**
- Doors — **UNLATCHED before touchdown**
- Evacuation — **AFTER STOPPING – emergency exits, upwind**

Engine Fire in Flight

FIRE warning (CAWS) and/or smoke, flames, ITT rise

LAND IMMEDIATELY

- Condition lever (memory) — **CUTOFF (propeller FEATHERS)**
- Fuel emergency shutoff (memory) — **CLOSED**
- Fuel pumps (memory) — **OFF**
- ECS / bleed air (memory) — **OFF (keep engine air out of the cabin)**
- Cabin pressurisation — **DUMP (when below 10,000 ft cabin) – ram air / vents as required**
- Oxygen masks — **DON 100% if smoke in cabin**
- Airspeed — **110–115 KIAS (increase if fire persists – sideslip to keep flames away)**
- Forced landing — **EXECUTE – Engine Failure in Flight forced-landing items; do not attempt restart**

No engine fire extinguisher is installed. Land on the nearest suitable surface; evacuate immediately after stopping.

Engine Fire on the Ground / During Start

FIRE warning, flames, or ITT rising rapidly past 1,000°C during start

- Condition lever (memory) — **CUTOFF**
- Fuel emergency shutoff (memory) — **CLOSED**
- Starter (memory) — **CONTINUE MOTORING to purge fuel / lower ITT (observe starter limits)**
- Fuel pumps — **OFF**

▶ Fire out?

Yes:

- Starter — **OFF**
- Battery 1 / 2 / external power — **OFF / DISCONNECTED**

Do not restart. Maintenance inspection required (hot section, inertial separator, cowling).

No:

- Starter — **OFF**
- Battery 1 / 2 — **OFF**
- Aircraft — **EVACUATE through the cabin / cargo door, away from the fire; ground extinguisher at the engine inlet / exhaust**

Emergency Descent

- PCL (memory) — **IDLE**
- Condition lever (memory) — **FLIGHT IDLE (propeller MAX RPM – drag)**

▶ Descent configuration?

Gear DOWN (maximum rate, below 180 KIAS):

- Landing gear (memory) — **DOWN (below 180 KIAS)**
- Flaps (memory) — **UP**
- Airspeed (memory) — **180 KIAS (Vle)**

Clean (smooth air, Vmo):

- Landing gear / flaps (memory) — **UP**
- Airspeed (memory) — **236 KIAS Vmo (165 KIAS Va in turbulence)**

- Autopilot / yaw damper — **OFF (or AP in VS / IAS mode if preferred)**
- Oxygen — **CREW MASKS ON above 10,000 ft cabin; passenger oxygen as required**
- ATC — **ADVISE – emergency descent, 7700 as required**
- Level-off — **10,000 ft or MEA – power restored smoothly, gear UP below 180 KIAS if continuing**

Rapid Decompression / Cabin Altitude High

CAB PRESS warning (cabin above 10,000 ft), loud decompression, or passenger masks dropped (approx 13,500 ft cabin)

LAND AS SOON AS PRACTICAL

- Crew oxygen masks (memory) — **DON – 100%; mic to OXY MASK**
- Passenger oxygen (memory) — **ON / MANUAL DEPLOY if masks have not dropped**
- Emergency descent (memory) — **INITIATE to 10,000 ft or MEA**

- Crew / passenger communication — **ESTABLISH; check passengers on oxygen**
- Cabin pressurisation — **CHECK (switch NORMAL not DUMP, bleed air ON, ECS AUTO, controller setting) – see Pressurisation Fault**
- ATC — **ADVISE**

▶ Cabin pressurisation recovered?

Yes:

- Cabin altitude — **MONITOR – stay at an altitude the cabin can hold below 10,000 ft**
- Deferred: Pressurisation fault – fly low, oxygen used, maintenance inspection. — **NOTED**

No:

- Altitude — **REMAIN at or below 10,000 ft (oxygen supply as required)**
- Deferred: Unpressurised – remain at or below 10,000 ft, oxygen quantity limited. — **NOTED**

Smoke / Fumes / Electrical Fire

LAND AS SOON AS POSSIBLE

Crew oxygen masks (memory) ..... **DON – 100% / EMER; mic to OXY MASK**

Smoke goggles (memory) ..... **ON (if carried)**

▶ **Source of smoke?**

Electrical:

Affected equipment / circuit breaker ..... **OFF / PULL (if identified)**

▶ **Source identified and smoke clearing?**

Yes:

Deferred: Electrical fire – affected circuit isolated, equipment unavailable. ..... **NOTED**

No:

Generator 1 / 2 ..... **OFF (battery power only)**

Non-essential bus / Avionics 2 ..... **OFF – essential bus loads only**

Hand fire extinguisher ..... **USE as required; ventilate afterwards**

Electrical equipment ..... **RESTORE one item at a time, essential only, checking for smoke; generators back ON one at a time if clear**

Landing gear is electro-hydraulic and flaps are electric: on battery only, extend the gear early (emergency hand pump available) and plan a flaps-up or flaps 15 landing.

Deferred: Electrical fire – battery / essential bus only, limited endurance (approx 30 min), early or manual gear extension, reduced flap landing. ..... **NOTED**

ECS / bleed air (oily fumes, haze from vents):

Bleed air ..... **OFF; ECS OFF**

Cabin pressurisation ..... **Cabin will depressurise – descend to keep cabin below 10,000 ft**

Ram air / vents ..... **OPEN to ventilate (below 10,000 ft)**

Deferred: Bleed air isolated – unpressurised, remain at or below 10,000 ft, maintenance (engine oil seal) inspection. ..... **NOTED**

Unknown:

Both isolations ..... **APPLY – electrical load shed and bleed air OFF**

Smoke / fume evacuation:

Descent ..... **TO 10,000 ft or below if practicable**

Cabin pressurisation ..... **DUMP (below 10,000 ft) to ventilate**

Fresh air / ram air ..... **MAXIMUM (bleed air ON if not the source)**

Passenger oxygen ..... **CONSIDER (masks do not protect from smoke)**

Stick Pusher / Stall Warning Malfunction

PUSHER caution, uncommanded pusher activation, or shaker / pusher not working as expected

LAND AS SOON AS PRACTICAL

Pusher interrupt (control wheel) (memory) ..... **PRESS & HOLD (uncommanded push)**

Control column (memory) ..... **HOLD / RECOVER – airspeed above 1.3 Vs**

Stick pusher / stall warning circuit breaker ..... **PULL (if uncommanded activation persists)**

Autopilot ..... **OFF if trim / pitch excursion**

▶ **Pusher system condition?**

Uncommanded activation – pusher disabled:

Airspeed ..... **MAINTAIN Vref +10 KIAS minimum on approach; no steep turns at low speed**

Icing conditions ..... **AVOID (no pusher ICE MODE protection)**

Deferred: Stick pusher disabled – no stall protection; Vref +10 KIAS, avoid icing, flaps 15 or 40 as briefed. ..... **NOTED**

PUSHER caution – system failed, no activation:

Stall warning ..... **ASSUME INOPERATIVE – fly by airspeed; avoid icing**

Deferred: Stall warning / pusher inoperative – Vref +10 KIAS, avoid icing conditions. ..... **NOTED**

Trim Runaway

Uncommanded pitch / rudder / aileron trim movement, A/P TRIM warning

Control column / rudder (memory) ..... **HOLD against the trim**

Trim interrupt / A/P disconnect (control wheel) (memory) ..... **PRESS & HOLD**

Autopilot (memory) ..... **OFF**

Affected trim circuit breaker (PITCH / RUDDER / AILERON TRIM) ..... **PULL**

Trim interrupt ..... **RELEASE – check trim stopped**

▶ **Affected axis?**

Pitch:

Manual pitch trim wheel ..... **TRIM to relieve forces (electric pitch trim unavailable)**

Airspeed ..... **Reduce to approach speeds early; avoid large configuration changes**

Deferred: Electric pitch trim inoperative – manual trim wheel only, no autopilot; approach briefed. ..... **NOTED**

Rudder / aileron:

Trim ..... **LEAVE as is – counter with controls; power / speed to minimise forces**

Deferred: Rudder / aileron trim inoperative – out-of-trim landing, no autopilot. ..... **NOTED**

Landing Gear Emergency Extension

Gear does not extend normally / HYDR caution / no three green

Airspeed	**BELOW 180 KIAS (approx 130 KIAS eases pumping)**
Landing gear hydraulic pump circuit breaker	**PULL**
Landing gear selector	**DOWN**
Emergency gear extension hand pump (between seats)	**HANDLE OUT – PUMP until 3 GREEN, then until firm resistance (approx 25–40 strokes – verify)**
Landing gear indicator	**3 GREEN, gear unsafe light out**

Do not reset the hydraulic pump circuit breaker or move the gear selector in flight after an emergency extension. Leave the gear down.

▶ 3 green obtained?

Yes:

Deferred: Gear extended by hand pump – do not retract; normal landing, gear system inspection required. ..... **NOTED**

No:

Use the Gear-Up / Gear Unsafe Landing procedure.

Deferred: Gear unsafe – gear-up / partial-gear landing, emergency services, evacuate after stopping. ..... **NOTED**

Gear-Up / Gear Unsafe Landing

Gear cannot be extended or indication unsafe after emergency extension

Runway	**LONGEST available, into wind; emergency services advised; burn off fuel if practical**
Passengers	**BRIEF – belts tight, brace position, exits**
Landing gear	**UP (or as it is – do not recycle) for a full gear-up landing**
Flaps	**40 (final approach)**
Airspeed	**Vref 80–90 KIAS – minimum speed at touchdown, shallow flare**
Cabin pressurisation	**DUMP; ECS / bleed air OFF**
Doors	**UNLATCHED on short final**
Condition lever	**CUTOFF on short final once landing assured (propeller feathers – reduces prop damage)**
Fuel emergency shutoff / fuel pumps	**CLOSED / OFF after touchdown assured**
Battery 1 / 2 / generators	**OFF after touchdown**
Evacuation	**AFTER STOPPING – upwind, clear of the aircraft**

Deferred: Gear-up landing – airfield closure, emergency services, evacuate after stopping. ..... **NOTED**

Propeller Overspeed / Governor Failure

Np above 1,700 RPM and not responding (PROP LOW PITCH may show), noise / vibration

PCL (memory)	**REDUCE to bring Np below 1,700 RPM**
Airspeed (memory)	**REDUCE (pitch up)**

▶ Np controllable at or below 1,700 RPM (overspeed governor holds approx 1,800 RPM)?

Yes:

Power	**SET to keep Np within limits (1,700 max; 1,870 transient 20 sec – verify)**
Altitude / airspeed	**Choose a combination that limits Np; avoid high-speed descents**

Deferred: Primary governor fault – Np controlled with PCL and airspeed, land as soon as practical. ..... **NOTED**

No – Np uncontrollable above transient limit:

Condition lever	**CUTOFF (propeller FEATHERS) – Engine Failure in Flight**

Deferred: Engine shut down for prop overspeed – forced landing / glide, land as soon as possible. ..... **NOTED**
